📘 **AI/ML Internship – Day 32 Notes & Tasks**

🟢 **Module 6: Seaborn Introduction & Statistical Visualization**

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📌 **Part 1: What is Seaborn?**

🔹 **Definition**

👉 Seaborn is a Python data visualization library built on top of Matplotlib.

👉 It is used for:

- Statistical visualization
- Attractive graphs
- Easy data analysis

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📌 **Part 2: Why Seaborn is Important?**

👉 Seaborn provides:

- Better-looking graphs
- Simple syntax
- Statistical charts

👉 Widely used in:

- Data Science
- AI/ML
- Analytics

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📌 **Part 3: Theory Answers (IMPORTANT)**

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🔹 **1. What is Seaborn?**

👉 Seaborn is a Python library used for advanced statistical visualization.

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🔹 **2. Difference Between Matplotlib and Seaborn**

| Matplotlib | Seaborn |
| --- | --- |
| Basic visualization | Advanced visualization |
| More coding | Simpler syntax |
| Simple design | Better design |

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### 🔹 3. Why Seaborn is Used in AI?

👉 Seaborn helps:

- Analyze datasets
- Visualize relationships
- Detect patterns

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### 🔹 4. What is Statistical Visualization?

👉 Visualization used for analyzing statistical patterns in data.

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### 🔹 5. What is a Distribution Plot?

👉 A graph showing how data values are distributed.

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## 📌 Part 4: Installing Seaborn

```
pip install seaborn
```

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## 📌 Part 5: Import Seaborn

```
import seaborn as sns
import matplotlib.pyplot as plt
```

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## 📌 Part 6: Simple Line Plot

```
import seaborn as sns
import matplotlib.pyplot as plt
```

```
x = [1, 2, 3, 4]
y = [10, 20, 30, 40]

sns.lineplot(x=x, y=y)

plt.show()
```

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📌 **Part 7: Bar Plot**

```
students = ["A", "B", "C"]
marks = [85, 90, 78]

sns.barplot(x=students, y=marks)

plt.show()
```

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👉 Used for comparison.

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📌 **Part 8: Scatter Plot**

```
x = [1, 2, 3, 4]
y = [5, 10, 15, 20]

sns.scatterplot(x=x, y=y)

plt.show()
```

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👉 Used to find relationships between variables.

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📌 **Part 9: Histogram**

```
data = [10, 20, 20, 30, 40, 40, 50]

sns.histplot(data)

plt.show()
```

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👉 Shows frequency distribution.

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### 📌 Part 10: Box Plot

🔹 **Important Statistical Graph**

```
data = [10, 20, 30, 40, 100]

sns.boxplot(x=data)

plt.show()
```

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👉 Helps identify:

- Outliers
- Spread of data
- Median values

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### 📌 Part 11: Count Plot

```
courses = ["AI", "Web", "AI", "Cloud", "AI"]

sns.countplot(x=courses)

plt.show()
```

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👉 Counts repeated categories.

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### 📌 Part 12: Styling with Seaborn

```
sns.set_style("darkgrid")

sns.lineplot(x=x, y=y)

plt.show()
```

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🔹 **Different Styles**

sns.set_style("whitegrid")
sns.set_style("dark")
sns.set_style("ticks")

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## 📌 Part 13: Real-World Thinking

👉 In AI/Data Science:

- Analyze model results
- Detect outliers
- Understand distributions
- Compare categories

👉 Seaborn is heavily used in:

- Exploratory Data Analysis (EDA)
- Data cleaning
- Statistical analysis

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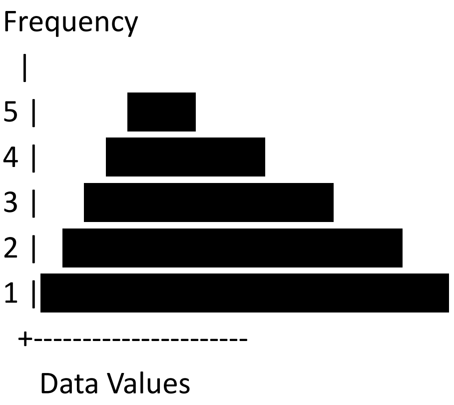
## 📌 Part 14: Visualization Example

Histogram

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## 🎯 Tasks for Day 32

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### 🧠 Task 1: Theory (Write Answers)

1. What is Seaborn?
2. Difference between Matplotlib and Seaborn
3. Why Seaborn is important?
4. What is statistical visualization?
5. What is box plot?

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### 💻 Task 2: Line Plot Practice

1. Create line graph
2. Add style
3. Compare data trends

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### 💻 Task 3: Bar Plot Practice

1. Student marks comparison
2. Sales comparison
3. Employee salary comparison

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### 💻 Task 4: Scatter Plot Practice

1. Height vs weight
2. Study hours vs marks
3. Temperature vs sales

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### 💻 Task 5: Histogram Practice

1. Create histogram
2. Analyze frequency
3. Use random dataset

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### 💻 Task 6: Box Plot Practice

1. Detect outliers
2. Analyze salary dataset
3. Analyze marks dataset

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### 💻 Task 7: Count Plot Practice

1. Course count
2. Department count
3. Product category count